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Fairmarkit: AI Autopilot for Procurement Orders

Final Report

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1 Fairmarkit: AI Autopilot for Reverse Auctions

Founded in Boston in 2017, Fairmarkit focuses on the long-tail of purchasing—i.e., the many low-value buys that collectively account for $\sim 20\%$ of spend and involve $\sim 80\%$ of suppliers. This analysis relies on the *Pareto principle*: it highlights a split between the small “head” of strategic, high-value transactions (roughly 20% of suppliers/contracts that drive $\sim 80\%$ of spend) and the complementary long tail [1]. Fairmarkit automates the long tail so procurement teams can concentrate their time on the most important, head-of-the-distribution deals. This portion of spend is particularly critical: optimizing the tail can yield material reductions in total procurement outlays—on the order of 5–10% [2].

The platform operates in two steps [3].

- *Intake*. It automatically converts eligible requests into compact, competitive *Requests for Quotation* (RFQs) (for fixed specifications) or *Requests for Proposal* (RFPs) (when the outcome/service is not fully specified). Once submitted, the platform recommends proposals from the platform’s supplier base and, where useful, extends sourcing to external pools (e.g., Amazon Business) to widen price discovery without compromising quality [4].
- *Negotiation*. The customer selects a negotiation mode—reverse auction or AI-driven single negotiation—and sets parameters (e.g., threshold bid and terms). The platform then runs the competition and—once the final offers are set—the client can accept the negotiation and finalize the request, while the platform writes awards back into the buyer’s Enterprise Resource Planning (ERP) stack.

2 Value Creation/Competitive Advantage

The value for the customers is twofold.

First, it frees teams to **focus on high-impact negotiations**. Second, for low-value, frequent buys it delivers a claimed average savings of about 11% versus pre-negotiated purchasing-group discounts (Table 1).

TABLE 1
Category savings ranges (low–high and average).

Category	Low	High	Average	Category	Low	High	Average
Agency	8%	12%	10.0%	Janitorial Supplies	5%	10%	7.5%
Advertising	7%	15%	11.0%	Machinery and Custom Parts	3%	9%	6.0%
Audio/Visual Equipment	8%	12%	10.0%	Mail	2%	5%	3.5%
Computer Equipment	7%	15%	11.0%	Maintenance Services	5%	15%	10.0%
Computer Peripherals	7%	13%	10.0%	MRO	6%	17%	11.5%
Creative Services	8%	20%	14.0%	Office Supplies	6%	13%	9.5%
Datacenter Goods	5%	20%	12.5%	Print	7%	12%	9.5%
Engineered Parts	4%	9%	6.5%	Promotional	7%	12%	9.5%
Event Management	2%	15%	8.5%	Promotional Merchandise	5%	20%	12.5%
Facilities	7%	15%	11.0%	Safety Equipment	5%	15%	10.0%
Facilities Services	5%	30%	17.5%	Servers/Back Office Computing	7%	11%	9.0%
Freight	2%	5%	3.5%	Software and Renewals	2%	5%	3.5%
Furniture	10%	18%	14.0%	Telephony/Network Equipment	5%	12%	8.5%
IT Hardware	7%	15%	11.0%	Temporary Help	7%	20%	13.5%
IT Software	5%	8%	6.5%	Translation	10%	15%	12.5%
IT Consulting	12%	18%	15.0%				

Source: Fairmarkit Presentation Deck 2023 (attached).

These savings stem from **broader supplier network** that yields to greater competitive pressure and thus **cheaper services** (often the primary decision criterion in low-value purchases) and **lower search and administrative costs** for the customer.

In summary, the value proposition is a **simplified** procurement process: a single-entry platform for low-value goods and services that recommends and automatically identifies suitable suppliers—based on procurement history and trends, manufacturer and SKU information—to provide relevant supplier for each request, thus expanding competition and reducing search and administrative effort. Moreover, it provides autonomous sourcing (reverse auctions or AI-driven negotiation) to **compress cycle times**, ultimately delivering measurable savings (see Table 2).

TABLE 2
Cycle-time compression: Manual ERP/P2P vs Fairmarkit + ERP/P2P.

	Manual with ERP/P2P	Fairmarkit + ERP/P2P
1 transaction	60 minutes	3 minutes
1,000 transactions	1,000 hours	50 hours
3,454 transactions	3,454 hours (431 working days)	173 hours (21 working days)
Headcount to support (CoE)	10 FTEs	1 FTE

Source: Fairmarkit Presentation Deck 2023.

Vendor-reported example; P2P = Procure-to-Pay.

3 Fairmarkit's Competitive Strategy and Value Net

The strategic positioning of Fairmarkit could be properly understood by analyzing its ecosystem (Value Net see. 3.1) and the resulting competitive strategy (Vertical/Horizontal differentiation see. 3.2).

3.1 Value Net

Fairmarkit operates within a complex ecosystem. The key players can be identified through a Value Net analysis (see Figure 1).

Fairmarkit's **customers** include large, global companies like *Maxar*, *SDI*, *Sonoco*, *bp* [5] and *Emirates Flight Catering*. These organizations have highly complex procurement needs and Fairmarkit's platform solves their specific challenge of tail spend [6]. As these are large enterprise buyers that place millions mid-to-low value procurement orders, properly managing this spend category can yield to significant savings.

To deliver its service, Fairmarkit relies on key **suppliers** like the major cloud infrastructure providers *Amazon Web Services*, *Google Cloud* or *Microsoft Azure*. These platforms provide the essential, scalable and secure computing power, data storage and hosting required to run Fairmarkit's sophisticated AI models. However, a second, more critical supplier group is the specialized AI and engineering talent required to build and maintain the product. The scarcity of this talent gives this supplier group high bargaining power and represents a significant cost driver for the company.

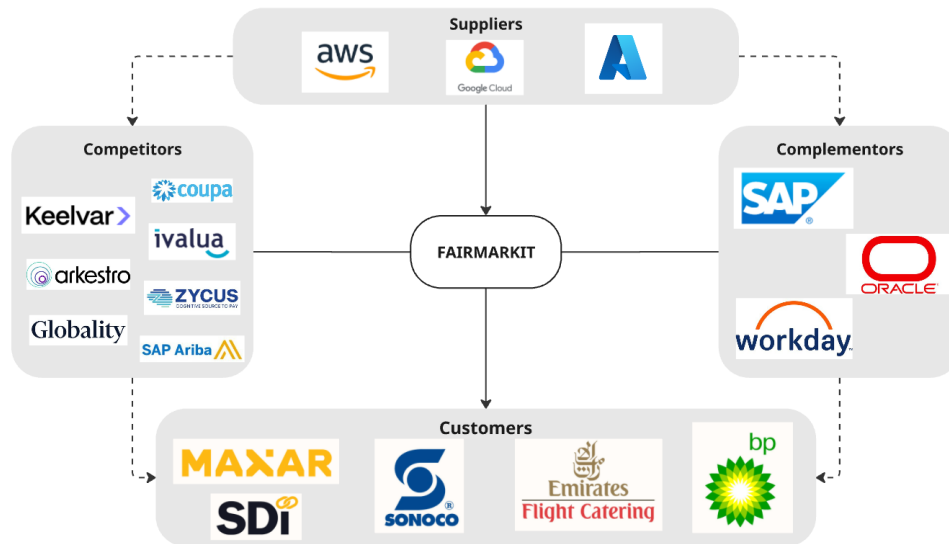


FIGURE 1
Fairmarkit's Brandenburger and Nalebuff's Value Net

Fairmarkit's **competitors** [7] fall into two broad groups. First, there are direct, niche rivals such as *Keelvar*, *Globality*, and *Arkestro*, which also develop AI-enabled sourcing tools [8]. These players typically position their products around (i) automation and optimisation of sourcing workflows, (ii) complex services spend, or (iii) predictive procurement orchestration, and they generally address procurement spend more broadly (i.e., beyond tail spend). This is precisely where Fairmarkit differentiates itself horizontally: it focuses specifically on automating the long-tail of purchasing for large enterprises (see 3.2). However, the rivalry in the segment is quite high as all are competing to be the best-in-class solution, leading to an arms race in AI features and innovation.

Second are the large, all-in-one Source-to-Pay procurement suites like *Coupa*, *Ivalua*, *Zycus*, and *SAP* [9]. These giants compete by offering a single, integrated platform that covers all procurement functions not just sourcing.

As such their main weakness is that their sourcing tools are often less agile or powerful than Fairmarkit's specialized solution. However, the greatest threat of a substitute comes not from another software, but from the status quo: a company's existing manual process of using emails and Excel [10]. Fairmarkit's primary challenge is proving its return on investment (ROI), first against this seemingly free alternative, and second against the good enough solution of the existing ERP.

However, Fairmarkit can rely on many **complementors**, as its platform is designed to be integrated into numerous ERPs. Core enterprise systems from *SAP*, *Oracle*, and *Workday* are crucial complementors [11]. They are vital because Fairmarkit is not designed to replace a company's entire financial system; rather, its value is unlocked by integrating with these systems. This deep integration is a core part of the strategy: it not only creates high switching costs for customers (reducing their bargaining power), but it also erects significant barriers to entry for any new competitor who would need to replicate these complex technical partnerships. The typical workflow involves purchase request starting in a complementor's system (like SAP), which FairMarkit's platform then intercepts to run its automated sourcing process. Once the best bid is secured, FairMarkit sends the final, cost-saving purchase order back into that same system for payment, acting as a powerful enhancement rather than a disruptive replacement.

A key insight into Fairmarkit's ecosystem is that some companies, like *SAP* and *Coupa*, lie in both the competitor and complementor categories and thus can be defined as "*co-opetitors*". This highlights Fairmarkit's core strategy: instead of trying to replace these giants, it coexists by positioning itself as a best-in-class add-on [12]. As such, these co-opetitors intensify the competitive rivalry, putting pressure on Fairmarkit to prove its superior ROI when compared to the large companies' good enough tools.

3.2 Vertical/Horizontal strategy

To differentiate itself against its competitors Fairmarkit's strategy is two-fold.

Firstly, Fairmarkit uses its **horizontal differentiation** to target the right customers. As such, Fairmarkit focuses on its core niche: organizations with a significant tail spend problem. This specific focus horizontally differentiates it from competitors like *Globality* which targets complex service spends. Furthermore, it targets customers who are interested and open to a best-in-class service i.e. those who prefer to plug specialized tools into their existing ERP systems rather than adopting the all-in-one suite philosophy.

Once it is in front of that ideal customer, it uses **vertical differentiation** to win the deal by proving its AI-driven automation is measurably superior to all other alternatives. Compared to the primary substitute—the status quo of using Excel and email—it is objectively faster, cutting cycle times from days to hours, and delivers a higher and more immediate return on investment (ROI) by unlocking over 10% in savings through competitive bidding. Against the good enough sourcing modules included in large incumbent suites like Coupa or SAP Ariba, Fairmarkit proves it is vertically superior by demonstrating its specialized AI is smarter, more autonomous, and ultimately more effective at finding savings.

4 Fairmarkit as an Intermediary

Fairmarkit's entire business model can be understood as a form of digital re-intermediation. Indeed Fairmarkit is an intermediary as it links and coordinates customers to their own suppliers and helps set prices guaranteeing quality and monitoring performance (see Figure 2).

As of today, Fairmarkit strategically positions itself as a new, more efficient intermediary, replacing two less effective alternatives.

The first is the traditional intermediary, such as a large-scale distributor, which simplifies purchasing but does so by adding opaque markups and eliminating price competition.

The second is total disintermediation, the internet-era promise of buyers going direct to suppliers, which in reality becomes an inefficient, manual nightmare of emails and spreadsheets for procurement teams.

Fairmarkit inserts itself as a smarter middleman; it is a technology platform, not a reseller. It doesn't hold inventory or add a hidden markup but rather acts as an electronic market-maker. It charges buyers a subscription fee to facilitate a transparent, competitive market for every purchase. Its AI automates the very work that made disintermediation so inefficient, allowing it to disintermediate the traditional intermediaries by stripping them of their pricing power. In this new model, a buyer can automatically get a quote from their old, preferred distributor while simultaneously forcing them to compete against a marketplace of other suppliers, shifting power from the reseller back to the buyer and the market itself.

In what follows, we first develop a quantitative model of Fairmarkit's interaction with its customers and derive the optimal two-part tariff. We then turn to the supply side of the intermediary relationship and model how many suppliers the platform should optimally contract with.

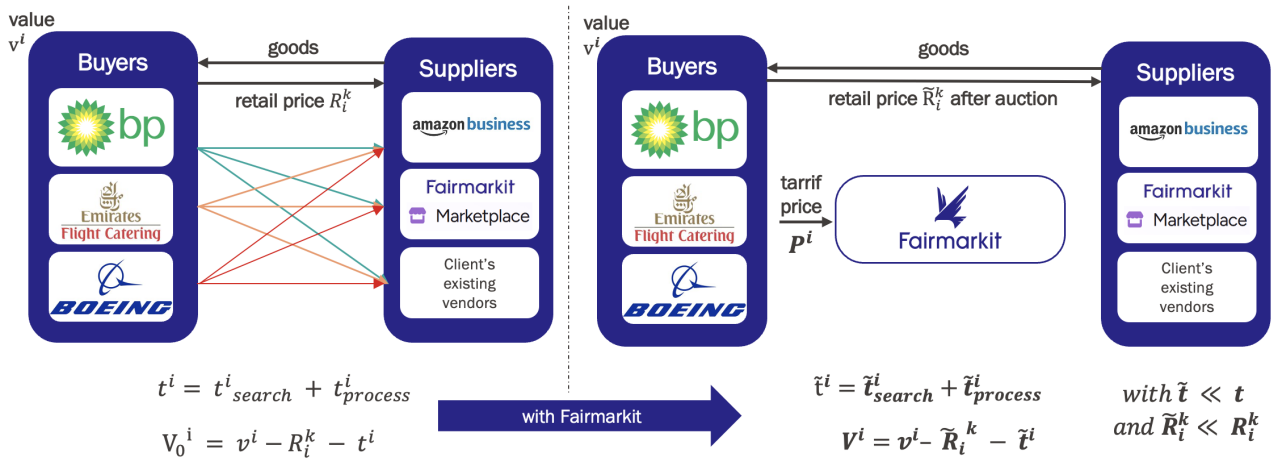


FIGURE 2

Fairmarkit as an Intermediary (analyzed from buyers' side)

5 Quantitative Framework

5.1 Pricing and Commercial Model

Fairmarkit is sold as an enterprise *SaaS* subscription: a multi-tenant, cloud-hosted application that integrates with ERP/P2P suites. Public materials indicate a subscription licensing model but do not disclose list prices; pricing is provided on request via the vendor or partner marketplaces [13, 14]. Typical enterprise contracts therefore combine (i) an annual platform subscription, typically scoped by modules and/or usage, and (ii) one-time implementation and integration services. This structure naturally maps into a two-part tariff (fixed fee plus usage price), which motivates the quantitative pricing framework developed below [15].

Set-up. Let n be the annual number of tail-spend requisitions routed to Fairmarkit (the adopted volume) and S the *average spend per requisition* (so total tail spend is nS). This n represents the customer's usage or adoption intensity, which does not necessarily correspond to the firm's total underlying procurement need. Let $\bar{h}$ be baseline processing and search time (in hours) per event, $\hat{h}$ the automated time ($\hat{h} \ll \bar{h}$), and w the fully-loaded hourly cost. Per vendor materials, *process and time savings* and *cost savings* are distinct drivers (see Sec. 2); thus s^{decl} denotes the vendor's **net price-savings rate on total tail spend**. We further specify the utility function to include a concave penalty term, parametrized by $\beta > 0$ that captures decreasing perceived benefits in the number of requisitions. Intuitively, as more transactions are routed through the automated procurement service, marginal improvements (in savings or process efficiency) decline, so the incremental value of additional requisitions diminishes.

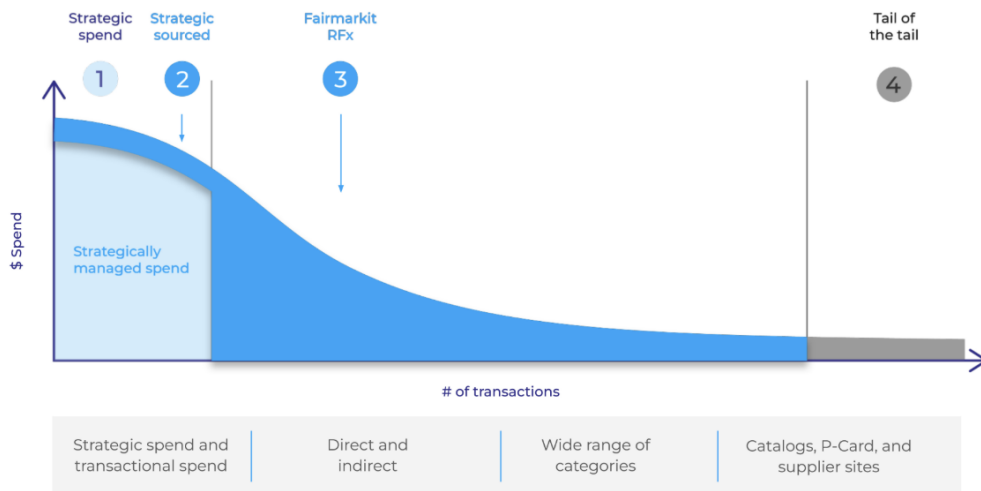


FIGURE 3

Long-tail of purchasing. Source: Fairmarkit website [16].

Thus, under the assumption of risk-neutral customers, we can model the *expected* incremental value versus an ERP/P2P baseline as:

$$\mathbb{E}[\Delta V] = \mathbb{E}[V - V_0] = \mathbb{E}[\text{WTP}] = \underbrace{n \mathbb{E}[S s^{\text{decl}}]}_{\text{price savings}} + \underbrace{n \mathbb{E}[w(\bar{h} - \tilde{h})]}_{\text{process + search savings}} - \underbrace{\frac{1}{2} \beta n^2}_{\text{concave disutility term}} \quad (1)$$

Pricing aligned to requisitions. Absent finer disclosures, we assume a two-part tariff (subscription + usage) with a possibly non-linear usage charge [17, 18]. We can therefore benchmark a feasible price band starting from the following formulation:

$$P_J = F + \phi(N_J), \quad P_J \leq \Delta V_J,$$

where F is the fixed subscription and $\phi(\cdot)$ is what we model in what follows to be a linear function of requisition volume. This establishes the baseline to assess pricing optimality against generated value.

Application. To operationalize this framework, we first make explicit how willingness to pay depends on requisition volume. The incremental value term ΔV in (1) is concave in the annual number of tail-spend requisitions n . Fairmarkit interacts with a heterogeneous population of customers that differ in their volume of indirect order requisitions. We capture this heterogeneity by introducing a one-dimensional type space

$$\Theta \subset \mathbb{R}_+, \quad \theta \sim g(\theta), \quad G(\theta),$$

where $g(\cdot)$ and $G(\cdot)$ denote the probability density function and cumulative distribution function, respectively, both assumed to be twice continuously differentiable on Θ ($g, G \in C^2$). Without loss of generality, we normalize the type space to $\Theta = [0, 1]$, so that θ can be interpreted as a unit-free index of value intensity or size (0 = least valuable customer, 1 = most valuable in the addressable market). In line with the value-creation mechanism described in Sec. 2 and consistent with Fairmarkit’s published customer case studies [19], we assume that the **distribution** of types is **shifted towards higher values**: firms with larger indirect spend and more complex procurement processes derive higher benefits from automating tail-spend sourcing, whereas very small firms face fewer requisitions and lower absolute savings, making adoption less attractive. Accordingly, the type distribution $g(\theta)$ on $[0, 1]$ is assumed to be left-skewed, with relatively more mass on high- θ customers in Fairmarkit’s addressable segment.

For a representative firm of type θ that routes n requisitions through the platform per year, we write its gross willingness to pay as

$$u(\theta, n) = \text{WTP}(\theta, n) = \theta n \alpha - \frac{1}{2} \beta n^2, \quad \alpha = \mathbb{E}[S s^{\text{decl}} + w(\bar{h} - \tilde{h})],$$

where α is the value created per requisition, combining net *price savings* on spend $S s^{\text{decl}}$ and *process and search savings* $w(\bar{h} - \tilde{h})$. As such, α is a reduced-form **compression of multi-dimensional value** into a single per-requisition parameter that the company could estimate through its clients. The parameter θ scales this per-requisition value and captures firm-level heterogeneity in value per requisition, while the concavity term is assumed common across firms as a reduced-form friction (e.g., in spend levels, internal cost of manual processing, or ability to exploit the tool).

Assuming that Fairmarkit adopts a two-part tariff composed of a fixed subscription component and a linear variable component, the total payment charged to a firm that generates n requisitions is therefore

$$P(n) = F + pn,$$

where F is the fixed fee and p is the per-requisition charge.

From an industry-structure perspective, Fairmarkit operates in a narrow niche of the broader procurement-automation market, focusing specifically on the tail-spend problem (see Sec. 3). Competing vendors offer AI-enabled sourcing or P2P automation, but typically do not target tail spend with the same degree of specialization, so that Fairmarkit is effectively differentiated on this dimension. Moreover, in many intermediary and platform markets, the provider that delivers the highest net value (for comparable costs) tends to attract a disproportionate share of transactions, generating de facto “winner-takes-most” outcomes.¹

Given this niche positioning and the tendency towards concentration in intermediary markets, it is a reasonable approximation to model Fairmarkit in isolation. In the pricing model below, we therefore abstract from strategic interactions with other vendors and treat the outside option (“ERP/P2P only” or a generic alternative tool) as exogenous. However, this is still a first-order approximation of the environment the company faces that could be

¹This follows the standard intuition from platform and intermediary markets, where cross-group network effects and multi-homing frictions can lead to highly concentrated market shares for the most valuable intermediary.

extended to cover also the interaction with the players in the market and how this may affect the company's pricing strategy.

Importantly, the tariff must lie below the incremental economic value generated for the customer, i.e. $P_J \leq \Delta V_J$ as defined above.

On the cost side, Fairmarkit relies on cloud infrastructure and compute capacity purchased in blocks. Such contracts typically feature stepwise capacity tiers with near-zero marginal cost for additional usage within a tier. We therefore model the cost function as piecewise constant and, on the relevant interior of a given capacity block, approximate the marginal cost of an extra requisition by $MC \approx 0$. This assumption is consistent with standard analyses of software and information goods, where fixed development and infrastructure costs dominate variable costs per transaction [20, 21].

Two-part tariff with heterogeneous customers. Given the heterogeneous type distribution $g(\theta)$, Fairmarkit chooses a two-part tariff (F, p) anticipating customer behaviour. A firm of type θ that faces the tariff (F, p) and chooses usage n obtains net utility

$$U(\theta, n; F, p) = \theta n \alpha - \frac{1}{2} \beta n^2 - (F + p n), \quad (2)$$

For given (F, p) , each type θ chooses its optimal requisition volume

$$n^*(\theta; p) \in \arg \max_{n \geq 0} U(\theta, n; F, p), \quad (3)$$

which induces an indirect utility $U(\theta; F, p) := U(\theta, n^*(\theta; p); F, p)$.

Types participate if and only if their indirect utility is non-negative. Define the participation set

$$\Theta_p(F, p) := \{\theta \in \Theta : U(\theta; F, p) \geq 0\}.$$

Equivalently, IR holds for all participating types:

$$U(\theta; F, p) \geq 0 \quad \forall \theta \in \Theta_p(F, p).$$

Under these assumptions, Fairmarkit's problem (initially abstracting from fixed infrastructure cost at the tier level which is sunk cost that does not impact in the optimization formulation) can be written as:

$$\max_{F, p} \mathbb{E}_\theta [F + p n^*(\theta; p)]$$

subject to the IR constraint above and the implied participation set of types. This formulation provides the basis for characterizing how the optimal two-part tariff (F, p) trades off rent extraction via the fixed fee against usage incentives via the per-requisition price in the presence of heterogeneous customers.

Solution of the two-part tariff problem

Following [15], we solve the two-part tariff problem in two steps: (i) characterize customer behaviour given a tariff (F, p) , and (ii) choose (F, p) to maximize expected profit subject to individual rationality.

Customer problem and best response. The first-order condition to solve (3) for an interior optimum is

$$\frac{\partial U}{\partial n}(\theta, n; F, p) = \theta \alpha - \beta n - p = 0, \implies n^*(\theta; p) = \max\left\{0, \frac{\theta \alpha - p}{\beta}\right\}.$$

In what follows, we focus on the interior region in which $\theta \alpha > p$ for all participating types, so that

$$n^*(\theta; p) = \frac{\theta \alpha - p}{\beta}. \quad (4)$$

Substituting (4) into (2), the indirect utility of type θ under tariff (F, p) is

$$\begin{aligned} U(\theta; F, p) &:= U(\theta, n^*(\theta; p); F, p) \\ &= \theta \alpha \frac{\theta \alpha - p}{\beta} - \frac{1}{2} \beta \left(\frac{\theta \alpha - p}{\beta} \right)^2 - \left(F + p \frac{\theta \alpha - p}{\beta} \right) \\ &= \frac{(\theta \alpha - p)^2}{2\beta} - F. \end{aligned} \quad (5)$$

The first term in (5) is the gross surplus (willingness to pay) of type θ at the optimal usage level, which we denote by

$$\tilde{U}(\theta, p) := \frac{(\theta \alpha - p)^2}{2\beta}.$$

Individual rationality and the marginal participating type. The tariff (F, p) must satisfy *individual rationality* (IR): only types with non-negative indirect utility participate. Formally,

$$U(\theta; F, p) = \tilde{U}(\theta, p) - F \geq 0 \quad \forall \theta \in [\theta_0, 1]. \quad (6)$$

Therefore, there exists a cutoff type $\theta_0 \in [0, 1]$ such that all types $\theta \geq \theta_0$ participate and types $\theta < \theta_0$ do not. The IR constraint of the marginal type θ_0 binds at the optimum:

$$U(\theta_0; F, p) = 0 \implies F = \tilde{U}(\theta_0, p) = \frac{(\theta_0 \alpha - p)^2}{2\beta}. \quad (7)$$

Expected profit and the optimal per-requisition price. Under the approximation of zero marginal cost per requisition (the cost is assumed to be piecewise constant K_j within a capacity tier), expected profit (normalized per potential customer) is equal to expected revenue:

$$\mathbb{E}[\Pi(F(\theta_0), p)] = \int_{\theta_0}^1 [F(\theta_0) + p n^*(\theta; p)] g(\theta) d\theta - K_j. \quad (8)$$

Using (4) and (7) (and expressing $\mathbb{E}[\Pi(F, p)]$ as Π for ease of notation), we can express Π as a function of p and the cutoff θ_0 :

$$\begin{aligned} \Pi(p, \theta_0) &= \int_{\theta_0}^1 \left[\frac{(\theta_0 \alpha - p)^2}{2\beta} + p \frac{\theta \alpha - p}{\beta} \right] g(\theta) d\theta - K_j \\ &= \underbrace{\frac{(\theta_0 \alpha - p)^2}{2\beta} [1 - G(\theta_0)]}_{\text{fixed-fee revenue}} + \underbrace{\frac{p}{\beta} \int_{\theta_0}^1 (\theta \alpha - p) g(\theta) d\theta}_{\text{usage-based revenue}} - K_j. \end{aligned} \quad (9)$$

Define the conditional mean type among participants,

$$\mu(\theta_0) := \mathbb{E}[\theta \mid \theta \geq \theta_0] = \frac{\int_{\theta_0}^1 \theta g(\theta) d\theta}{\int_{\theta_0}^1 g(\theta) d\theta},$$

so that $\mu(\theta_0) \in [\theta_0, 1]$. Then

$$\int_{\theta_0}^1 (\theta \alpha - p) g(\theta) d\theta = [1 - G(\theta_0)] (\alpha \mu(\theta_0) - p),$$

and (9) simplifies to

$$\Pi(p, \theta_0) = [1 - G(\theta_0)] \left\{ \frac{(\theta_0 \alpha - p)^2}{2\beta} + \frac{p}{\beta} (\alpha \mu(\theta_0) - p) \right\} - K_j. \quad (10)$$

Thus, Fairmarkit chooses the induced cutoff (and hence the share of total demand to be covered) and the per-requisition price by solving the joint problem

$$\max_{p, \theta_0} \Pi(p, \theta_0).$$

Best response in p for a given cutoff. For a given cutoff θ_0 the feasible set for the marginal price is $0 \leq p \leq \theta_0 \alpha$, so that usage of the marginal type remains non-negative. Using $\beta > 0$, we can drop the common positive multiplicative constant and differentiate the bracketed term in (10) with respect to p . Define

$$f(p; \theta_0) := \frac{(\theta_0 \alpha - p)^2}{2} + p(\alpha \mu(\theta_0) - p),$$

so that $\Pi(p, \theta_0) \propto [1 - G(\theta_0)] f(p; \theta_0)$. Then

$$\frac{\partial \Pi}{\partial p}(p, \theta_0) \propto \frac{\partial f}{\partial p}(p; \theta_0) = \alpha(\mu(\theta_0) - \theta_0) - p = 0 \implies p_{\text{unc}}^*(\theta_0) = \alpha(\mu(\theta_0) - \theta_0). \quad (11)$$

The second derivative is $\frac{\partial^2 \Pi}{\partial p^2}(p, \theta_0) \propto \frac{\partial^2 f}{\partial p^2}(p; \theta_0) = -1 < 0$, so profit is strictly concave in p . Imposing the feasibility constraint $0 \leq p \leq \theta_0 \alpha$, the constrained best response is therefore

$$p^*(\theta_0) = \min\{\theta_0 \alpha, \max\{0, \alpha(\mu(\theta_0) - \theta_0)\}\}. \quad (12)$$

In what follows we focus on the interior region where $0 < \alpha(\mu(\theta_0) - \theta_0) < \theta_0 \alpha \iff \theta_0 < \mu(\theta_0) < 2\theta_0$, so that

$$p^*(\theta_0) = \alpha(\mu(\theta_0) - \theta_0)$$

lies strictly inside the feasible set.

Cutoff choice with interior $p^*(\theta_0)$. Substituting the interior best response $p^*(\theta_0) = \alpha(\mu(\theta_0) - \theta_0)$ into (10), expected profit (up to the positive factor α^2/β and the constant K_j) can be written as

$$\Pi(\theta_0) \propto [1 - G(\theta_0)] \phi(\theta_0),$$

with

$$\phi(\theta_0) := \theta_0(\mu(\theta_0) - \theta_0) + \frac{(2\theta_0 - \mu(\theta_0))^2}{2} = \theta_0^2 - \theta_0\mu(\theta_0) + \frac{1}{2}\mu(\theta_0)^2. \quad (13)$$

Let $S(\theta) := 1 - G(\theta)$ denote the survival function and $g(\theta) := G'(\theta)$ the density.

For any $\theta \in (0, 1)$ by integration by parts we have:

$$\int_{\theta}^1 tg(t) dt = [tG(t)]_{\theta}^1 - \int_{\theta}^1 G(t) dt = \theta(1 - G(\theta)) + \int_{\theta}^1 (1 - G(t)) dt = \theta S(\theta) + \int_{\theta}^1 S(u) du,$$

Dividing by $S(\theta)$ gives the conditional mean

$$\mu(\theta) = \frac{\int_{\theta}^1 tg(t) dt}{S(\theta)} = \theta + \frac{1}{S(\theta)} \int_{\theta}^1 S(u) du.$$

Therefore, we can define the difference $\mu(\theta) - \theta$ as

$$B(\theta) := \mu(\theta) - \theta = \frac{1}{S(\theta)} \int_{\theta}^1 S(u) du \geq 0.$$

Using $\mu(\theta_0) = \theta_0 + B(\theta_0)$, equation (13) simplifies to

$$\phi(\theta_0) = \frac{1}{2}(\theta_0^2 + B(\theta_0)^2),$$

so, up to constants,

$$\Psi(\theta_0) := \Pi(\theta_0) \propto S(\theta_0) \phi(\theta_0) = \frac{1}{2}S(\theta_0)(\theta_0^2 + B(\theta_0)^2),$$

and the corresponding interior price is $p^*(\theta_0) = \alpha B(\theta_0)$.

To characterize interior extrema, differentiate $\Psi(\theta_0)$. From the representation of $B(\theta)$ above,

$$B'(\theta) = -1 + \frac{g(\theta)}{S(\theta)} B(\theta),$$

so

$$\phi'(\theta_0) = \theta_0 + B(\theta_0)B'(\theta_0) = \theta_0 - B(\theta_0) + \frac{g(\theta_0)}{S(\theta_0)} B(\theta_0)^2.$$

Using $S'(\theta_0) = -g(\theta_0)$, we obtain

$$\Psi'(\theta_0) = S'(\theta_0)\phi(\theta_0) + S(\theta_0)\phi'(\theta_0) = -g(\theta_0)\phi(\theta_0) + S(\theta_0)\phi'(\theta_0).$$

Substituting the expressions for ϕ and ϕ' and simplifying yields

$$\Psi'(\theta_0) \propto (B(\theta_0) - \theta_0) \left[g(\theta_0)(B(\theta_0) + \theta_0) - 2S(\theta_0) \right].$$

Recalling that $B(\theta_0) + \theta_0 = \mu(\theta_0)$ and defining the hazard rate $h(\theta_0) := g(\theta_0)/S(\theta_0)$, any interior extremum $\theta_0^* \in (0, 1)$ must therefore satisfy

$$B(\theta_0^*) = \theta_0^* \quad \text{or} \quad h(\theta_0^*) \mu(\theta_0^*) = 2.$$

The first case $B(\theta_0^*) = \theta_0^*$ is equivalent to $\mu(\theta_0^*) = 2\theta_0^*$, in which case

$$p^*(\theta_0^*) = \alpha(\mu(\theta_0^*) - \theta_0^*) = \theta_0^* \alpha,$$

so the unconstrained best response hits the upper bound of the feasible set for p and the price is not interior. Hence an interior optimum must satisfy

$$h(\theta_0^*) \mu(\theta_0^*) = 2, \quad \theta_0^* < \mu(\theta_0^*) < 2\theta_0^*, \quad (14)$$

where the inequalities ensure $0 < p^*(\theta_0^*) < \theta_0^* \alpha$.

In general, $\Psi(\theta_0)$ need not be concave in θ_0 , so (14) characterizes candidate interior optima rather than guaranteeing a unique global maximum. If the solution to (14) violates the feasibility inequalities $\theta_0^* < \mu(\theta_0^*) < 2\theta_0^*$, or if no solution exists, the optimum lies at a boundary in p . In that case the joint problem

$$\max_{\theta_0 \in (0,1), 0 \leq p \leq \theta_0 \alpha} \Pi(p, \theta_0)$$

is solved by comparing the two one-dimensional branches

$$\max_{\theta_0 \in (0,1)} \Pi(0, \theta_0) \quad \text{and} \quad \max_{\theta_0 \in (0,1)} \Pi(\theta_0 \alpha, \theta_0),$$

and choosing the best of these boundary solutions together with any feasible interior candidate.

Condition (14) has a natural **economic interpretation**. The hazard rate $h(\theta_0)$ measures the “instantaneous exit risk” at the cutoff (how quickly the tail of the distribution thins out), while $\mu(\theta_0)$ is the average type among all customers who decide to participate. At the optimal interior cutoff θ_0^* , the product of these two objects is pinned down at the constant 2. This balance condition captures **the trade-off between expanding coverage (lowering θ_0 increases the measure of customers served)** and preserving the average profitability of the served segment (higher θ_0 raises the average type and hence the fixed-fee rent extraction).

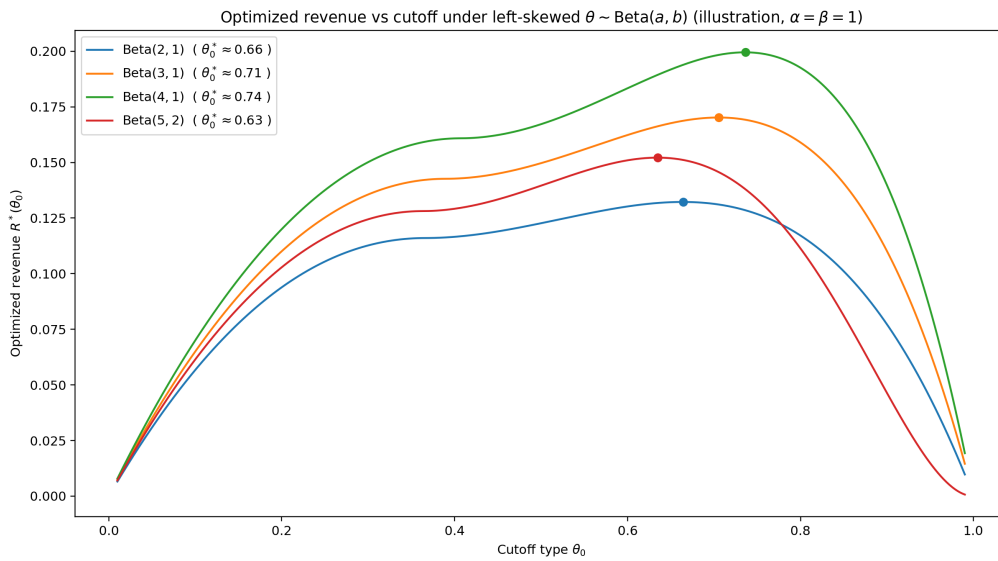


FIGURE 4

Optimized tariff revenue $R^*(\theta_0)$ versus cutoff θ_0 under $\theta \sim \text{Beta}(a, b)$ (illustration; $\alpha = \beta = 1$). With $MC \approx 0$, revenue maximization is equivalent to profit maximization, so we plot revenues for simplicity; for each θ_0 , $R^*(\theta_0)$ is obtained by maximizing over $p \in [0, \theta_0 \alpha]$ (interior candidate and boundary branches). Dots mark the optimal cutoff θ_0^* for each distribution.

Implications of piecewise-constant capacity costs. The baseline pricing result is derived under the within-tier approximation $MC \approx 0$, which makes profit smooth in p and yields a well-defined interior candidate p^{FOC} from the first-order condition.

In practice, infrastructure is purchased in discrete capacity blocks. Let total expected requisitions be

$$\mathcal{N}(p; \theta_0) := \int_{\theta_0}^1 n^*(\theta; p) g(\theta) d\theta = \int_{\theta_0}^1 \frac{\theta \alpha - p}{\beta} g(\theta) d\theta,$$

so capacity costs can be written as $C(\mathcal{N}(p; \theta_0))$ and tier thresholds are defined by $\mathcal{N}(p; \theta_0) \leq \bar{N}_1$, $\bar{N}_1 < \mathcal{N}(p; \theta_0) \leq \bar{N}_2$, etc.

Under this formulation and the assumption of discrete capacity blocks we model total cost as stepwise in usage:

$$C(\mathcal{N}) = \begin{cases} K_1 & 0 \leq \mathcal{N} \leq \bar{N}_1, \\ K_2 & \bar{N}_1 < \mathcal{N} \leq \bar{N}_2, \\ \vdots & \end{cases} \quad \text{with } K_{j+1} > K_j.$$

Thus increasing p reduces expected requisitions and can move the platform into cheaper capacity tiers. Equivalently, there exist price thresholds $p_1 > p_2 > \dots$ such that $\mathcal{N}(p) \leq \bar{N}_1$ iff $p \geq p_1$, $\bar{N}_1 < \mathcal{N}(p) \leq \bar{N}_2$ iff $p_2 \leq p < p_1$, etc.

Write the *gross* (pre-infrastructure) profit component as $\tilde{\Pi}(p)$, concave in p as we showed above. Then, within tier j ,

$$\Pi_j(p) = \tilde{\Pi}(p) - K_j,$$

so each tier induces the *same concave shape* in p , shifted downward by the fixed cost difference. Hence the global objective $\Pi(p)$ is only *piecewise* concave: the FOC identifies a *local* maximizer within a tier, but crossing a threshold causes a discrete drop in profit due to $K_j \rightarrow K_{j+1}$.

This yields two economically relevant cases:

- *Case 1 (FOC candidate in the lowest-cost tier)*. If $p^{FOC} \geq p_{n-1}$, the interior candidate is also the global optimum. Increasing p moves along a concave curve away from its peak, while decreasing p eventually forces $\mathcal{N}(p)$ above $\bar{N}_1$ and triggers a cost jump to K_2 , which strictly reduces profits.
- *Case 2 (FOC candidate in a higher-cost tier)* (see Fig. 5). If $p^{FOC} < p_{n-1}$, the interior maximizer could lie within a lower tier. The optimal policy must then compare this interior point to *threshold-avoiding boundary prices* that push the platform back into cheaper capacity blocks, thus increasing the price to the first one to enter the new tier n . Intuitively, even if raising p is locally suboptimal within the current tier (concavity), it may be globally optimal because the reduction in fixed infrastructure cost dominates the loss in gross profit.

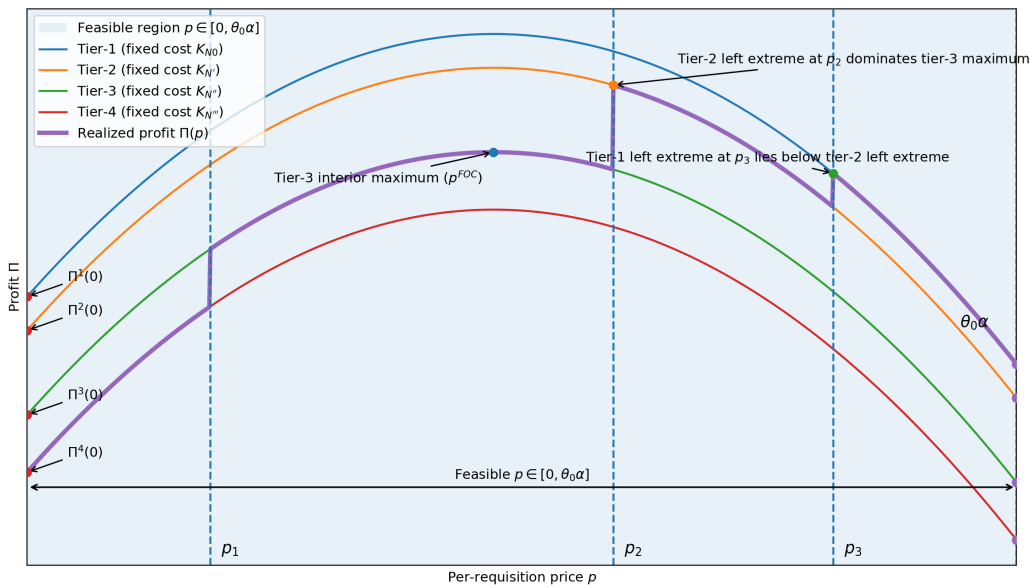


FIGURE 5

Schematic effect of capacity tiers on optimal per-requisition pricing. Within each capacity block, profit is concave in p (for a given threshold type θ_0) and admits a local FOC optimum; crossing a usage threshold induces a discrete cost increase, shifting profit downward. The global optimum can therefore occur at a threshold-avoiding boundary price rather than at the interior FOC solution.

Therefore, under piecewise-constant capacity costs, the optimal per-requisition price is generally the best of: (i) the within-tier interior solution, and (ii) boundary prices that keep usage just below a capacity threshold. This directly supports a strictly positive usage price as a demand-management instrument, even when within-tier marginal usage costs are near zero.

Illustration for a left-skewed Beta type distribution. Fairmarkit can estimate the type distribution $G(\theta)$ from platform data by using each customer's observed routed volume n_i under the current tariff and inverting the model-implied demand

$$n^*(\theta; p) = \left(\frac{\theta\alpha - p}{\beta} \right)_+ \Rightarrow \hat{\theta}_i = \frac{p_i + \beta n_i}{\alpha} \text{ for customers with } n_i > 0.$$

The empirical CDF of the implied types $\hat{G}(\theta) = \frac{1}{N} \sum_{i=1}^N \mathbf{1}\{\hat{\theta}_i \leq \theta\}$, provides an estimate of $G(\theta)$, while customers who never adopt can be treated as left-censored observations using funnel/pilot data.

Moreover, to make condition (14) more concrete, consider a left-skewed type distribution $\theta \sim \text{Beta}(2, 1)$ on $[0, 1]$, which places relatively more mass on high-value customers. In this case

$$g(\theta) = 2\theta, \quad G(\theta) = \theta^2, \quad S(\theta) = 1 - \theta^2, \quad \text{with hazard rate:} \quad h(\theta) = \frac{g(\theta)}{S(\theta)} = \frac{2\theta}{1 - \theta^2}.$$

The conditional mean among participants is

$$\mu(\theta_0) = \mathbb{E}[\theta \mid \theta \geq \theta_0] = \frac{\int_{\theta_0}^1 2t^2 dt}{1 - \theta_0^2} = \frac{2(1 - \theta_0^3)}{3(1 - \theta_0^2)}.$$

Using (14) to this Beta(2,1) case, an interior optimal cutoff $\theta_0^* \in (0, 1)$ must satisfy

$$h(\theta_0^*) \mu(\theta_0^*) = \frac{2\theta_0^*}{1 - (\theta_0^*)^2} \cdot \frac{2(1 - (\theta_0^*)^3)}{3(1 - (\theta_0^*)^2)} = 2,$$

i.e.

$$\frac{4\theta_0^*(1 - (\theta_0^*)^3)}{3(1 - (\theta_0^*)^2)^2} = 2.$$

Solving this equation numerically yields a unique solution

$$\theta_0^* \approx 0.66.$$

At this cutoff the conditional mean is

$$\mu(\theta_0^*) \approx 0.84,$$

so the interior price

$$p^*(\theta_0^*) = \alpha(\mu(\theta_0^*) - \theta_0^*) \approx 0.18 \alpha$$

indeed lies strictly inside the feasible interval $(0, \theta_0^* \alpha)$, and the feasibility conditions $\theta_0^* < \mu(\theta_0^*) < 2\theta_0^*$ are satisfied. In this illustrative left-skewed environment the platform optimally chooses a relatively high cutoff θ_0^* (serving only the upper third of the type support), and the hazard-mean condition $h(\theta_0^*)\mu(\theta_0^*) = 2$ pins down an interior per-requisition price $p^* > 0$: Fairmarkit screens out lower-value customers, extracting rent through a combination of a high fixed fee and a strictly positive usage charge.

Distributionally robust two-part tariff. The baseline analysis assumes that Fairmarkit can estimate the customer-type distribution from market data with sufficient precision to treat it as known. If, instead, Fairmarkit only observes an estimate $\hat{G}$ and faces misspecification risk, the design problem can be cast in distributionally robust form. Let $\mathcal{G}_\varepsilon$ denote an ambiguity set of distributions in a uniform Kolmogorov/KS neighborhood of $\hat{G}$ [22], for instance

$$\mathcal{G}_\varepsilon := \left\{ G : \sup_{\theta \in [0,1]} |G(\theta) - \hat{G}(\theta)| \leq \varepsilon \right\}.$$

The platform then chooses (θ_0, p) to maximize worst-case expected profit:

$$\max_{\theta_0 \in (0,1), 0 \leq p \leq \alpha \theta_0} \inf_{G \in \mathcal{G}_\varepsilon} \Pi_G(\theta_0, p), \quad G^{\text{wc}}(\theta_0, p) \in \arg \inf_{G \in \mathcal{G}_\varepsilon} \Pi_G(\theta_0, p).$$

Conditional on any fixed G , in regimes where the optimal variable price is interior, the best-response retains the same functional form but is evaluated under a least-favorable distribution:

$$p^{*,\text{rob}}(\theta_0) = \alpha(\mu_{G^{\text{wc}}}(\theta_0) - \theta_0), \quad \mu_{G^{\text{wc}}}(\theta_0) := \mathbb{E}_{G^{\text{wc}}(\theta_0, p^{*,\text{rob}}(\theta_0))}[\theta \mid \theta \geq \theta_0].$$

Likewise, the cutoff condition is obtained by replacing the distribution-dependent objects (e.g., tail mass and conditional moments) with their least favorable counterparts. Robustness therefore preserves the screening logic of the model, but yields a tariff and coverage choice that are conservative with respect to type-distribution uncertainty.

5.2 Reverse auction system

Formally, procurement auctions invert the classic seller-auction setting: *suppliers bid prices to win the buyer's contract* [23].

Assumptions.

- Each supplier $i \in \{1, \dots, m\}$ has a private cost $c_i \sim F$ on $[\underline{c}, \bar{c}]$ (IPC), with F continuous and strictly increasing (density f).
- Mid to Low-value requisitions that Fairmarkit automatizes are sufficiently standardized so that the award depends only on the bid b_i ; if quality matters, one can use a scoring/evaluated-cost rule [24].

- We adopt a **symmetric (i.i.d.) supplier benchmark** (F) because, within a narrowly defined tail-spend segment (fixed specification and pre-qualification), participating suppliers are *ex ante* comparable in capabilities and thus can be treated as draws from a common cost distribution.
- Bidders are risk-neutral and maximize expected profit.

The platform supports: *Dutch* (descending acceptance), *Japanese clock* (descending price with public drop-outs), and iterative undercutting, pay-as-bid *English* variant.

Pay-as-bid (first-price) formats. In *Dutch* and *English* reverse auctions, the Bayesian–Nash objective for supplier i to maximize is

$$\pi_i(b_i) = (b_i - c_i) \Pr\{b_i < b_j \forall j \neq i\},$$

with ties having zero probability under continuity. The same payoff structure applies to sealed-bid first-price procurement.

Second-price–equivalent formats. In *Japanese clock*, each supplier exits when price falls to cost ($b_i = c_i$ is weakly dominant). The last remaining supplier wins and pays the second-lowest drop-out price $p = c_{(2)}$, so

$$\pi_i^{2\text{nd}} = \mathbb{E}[(c_{(2)} - c_i) \mathbf{1}\{c_i = c_{(1)}\}].$$

Both formats are allocatively efficient in the symmetric equilibrium (lowest cost wins).

Remark. Under *symmetric* IPV with risk-neutral bidders, *revenue equivalence* implies that auction formats that implement the same allocation rule (lowest cost wins) and normalize the marginal type’s expected surplus in the same way generate the same *expected* procurement payment [25]. Hence, in this benchmark, Dutch/first-price and Japanese/second-price reverse auctions are **equivalent in terms of expected service cost** to the buyer, even though their bidding dynamics and price paths differ.

Once symmetry is relaxed to *asymmetric* IPV (independent but non-identical F_i), revenue equivalence generally fails: both **expected prices** and (in the optimal design problem) potentially the **allocation rule** can differ across first- vs. second-price mechanisms, and there is no distribution-free ranking of standard formats. In asymmetric environments, the revenue-optimal (cost-minimizing) mechanism is typically *Myersonian* and may require bidder-specific reserves or weights tailored to (F_i) rather than coinciding with any canonical format [26]. This provides a structural reason for why a **platform may retain multiple reverse-auction formats**: when supplier pools differ in composition across categories, the formats can deliver meaningfully different outcomes in both allocative performance and expected customer procurement cost.

Why format often matters less than participation (Bulow–Klemperer [27]). Under symmetric IPV with regular distributions, adding one extra bidder in a simple second-price/English auction yields (weakly) higher seller revenue than using the optimal auction with one fewer bidder [27]. In procurement (reverse auctions), the corollary flips signs: the buyer’s *expected procurement cost* with $m+1$ symmetric suppliers, a simple Vickrey/English reverse auction yields (weakly) lower expected procurement payment than any optimal IC mechanism with only m suppliers (and by revenue equivalence the comparison applies in expectation to standard pay-as-bid formats as well):

$$\mathbb{E}[\text{Cost}_{\text{simple}}^{m+1}] \leq \mathbb{E}[\text{Cost}_{\text{optimal}}^m] \leq \mathbb{E}[\text{Cost}_{\text{simple}}^m] \quad (15)$$

Consider supplier expansion in a *common* tail-spend **category** $\omega \in \Omega$ (see Table 1) that is routinely **purchased across the participating customer base**. Define the *within-category (per-requisition) participation gain* from adding one supplier in category $\omega \in \Omega$ under a standard reverse-auction format as

$$\delta_m^\omega := \mathbb{E}[\text{Cost}_{\text{simple}}^{m,\omega}] - \mathbb{E}[\text{Cost}_{\text{simple}}^{m+1,\omega}] \geq 0,$$

i.e. the reduction in the buyer’s *expected procurement payment per requisition of category* ω . By (15) applied at category level, this within-category participation gain is at least as large as the corresponding mechanism-design gain at fixed m .

In the pricing model (Sec. 5.1), Fairmarkit’s per-requisition value parameter is $\alpha = S s^{\text{decl}} + w(\bar{h} - \tilde{h})$, which aggregates price savings on spend and process savings. We capture the fact that the improvement applies only to requisitions in ω by assuming that, for all participating types $\theta \geq \theta_0^*$, a constant share $\lambda_\omega \in (0, 1)$ of their platform requisitions belongs to that category, i.e. $n_\omega(\theta) = \lambda_\omega n(\theta)$. Using the definition of δ_m^ω above, the effective per-requisition value over the customer’s entire requisition basket is the weighted average

$$\alpha' = (1 - \lambda_\omega)\alpha + \lambda_\omega(\alpha + \delta_m^\omega) = \alpha + \lambda_\omega \delta_m^\omega.$$

Since the optimal cutoff type θ_0^* is pinned down by the type distribution (hence does not depend on α in the interior solution), the incremental gain in fully optimized profit from expanding the supplier pool is

$$\Delta \Pi_m^*(\theta_0^*) := \Pi^*(\theta_0^*; \alpha') - \Pi^*(\theta_0^*; \alpha), \quad (16)$$

Closed form (interior-price regime). Let $S(\theta_0) := 1 - G(\theta_0)$ and $\mu(\theta_0) := \mathbb{E}[\theta \mid \theta \geq \theta_0]$. In the interior regime $\mu(\theta_0) \leq 2\theta_0$, the optimal marginal price is $p^*(\theta_0; \alpha) = \alpha(\mu(\theta_0) - \theta_0)$, and evaluating the integral in (8) at the optimal tariff yields

$$\Pi^*(\theta_0; \alpha) = \frac{S(\theta_0) \alpha^2}{2\beta} (\mu(\theta_0)^2 - 2\theta_0\mu(\theta_0) + 2\theta_0^2). \quad (17)$$

Therefore,

$$\Delta\Pi_m^*(\theta_0^*) = \frac{S(\theta_0^*)}{2\beta} (\alpha'^2 - \alpha^2) (\mu(\theta_0^*)^2 - 2\theta_0^*\mu(\theta_0^*) + 2(\theta_0^*)^2). \quad (18)$$

However, expanding the supplier base is not free. Let $T_{m,\omega}$ denote Fairmarkit's (annualized) search and onboarding cost of bringing the $(m+1)$ -st supplier into the relevant pool for category ω . The marginal expansion from m to $m+1$ suppliers is privately worthwhile up to the point where the incremental gain in fully optimized profit equals the marginal search cost:

$$\frac{S(\theta_0^*)}{2\beta} (\alpha'^2 - \alpha^2) (\mu(\theta_0^*)^2 - 2\theta_0^*\mu(\theta_0^*) + 2(\theta_0^*)^2) = T_{m,\omega}. \quad (19)$$

Under *revenue monotonicity* and *revenue submodularity*, expected auction revenue is increasing and exhibits diminishing marginal returns as the bidder set expands; in i.i.d. environments this corresponds to concavity of expected revenue in the number of bidders [28]. Using the standard procurement-to-auction transformation $v = \bar{c} - c$ and $\tilde{b} = \bar{c} - b$, supplier profits satisfy $v - \tilde{b} = b - c$. For a single-item first-price (pay-as-bid) procurement, the winning payment is $b_{(1)} = \bar{c} - \tilde{b}_{(m)}$, hence $\mathbb{E}[b_{(1)}] = \bar{c} - \mathbb{E}[\text{Rev}(m)]$.

Therefore, concavity of $\mathbb{E}[\text{Rev}(m)]$ implies **convexity** of the expected procurement payment in m , i.e., **diminishing marginal savings** from recruiting additional suppliers. Assuming the marginal onboarding/search cost $T_\omega(m)$ as increasing (e.g., $T_\omega(\cdot)$ convex in m because the most accessible suppliers are recruited first and the remaining ones are progressively harder to identify, qualify, and integrate), then the optimal supplier pool size is characterized by a standard marginal condition: recruit up to the point where the incremental profit gain (equivalently, the marginal expected savings) from adding the $(m+1)$ -st supplier equals $T_\omega(m)$; if the curves cross once, this yields a unique interior m^* , otherwise the optimum is at a boundary.

In summary, supplier recruitment raises effective value creation only through the *affected* requisition share λ_ω , shifting α to $\alpha' = \alpha + \lambda_\omega \delta_m^\omega$ and thereby increasing fully optimized platform profit under the two-part tariff. Recruitment for a specific procurement category should hence be pushed only up to the point where the **incremental gain in fully optimized profit** satisfies $\Delta\Pi_m^*(\theta_0^*) = T_{m,\omega}$.

5.3 Key Quantitative Takeaways

From the model of two-part tariff and the Reverse Auction system, one can derive the following strategic implications:

- **Customer Heterogeneity:** The **company-based** assumption of left-skewed type distribution ($g(\theta)$) on customer willingness to pay implies that Fairmarkit optimally targets high- θ enterprise segments, choosing a high cutoff θ_0^* and screening out lower-value customers. Indeed, the unconstrained interior optimal per-requisition price is positive: $p^* = \alpha(\mu(\theta_0) - \theta_0)$. This formula links the price to the difference between the average type among participants ($\mu(\theta_0)$) and the marginal cut-off type (θ_0).
- **The Hazard-Mean Tradeoff for Coverage:** The optimal cutoff θ_0^* (which determines the share of the market to be served) is governed by the condition $h(\theta_0^*)\mu(\theta_0^*) = 2$. This formalizes the tradeoff: lowering θ_0 expands coverage, while raising it increases the average profitability of the served segment.
- **Cost-Capacity Tiers:** With stepwise infrastructure costs $C(\mathcal{N}) = K_j$ for $\bar{N}_{j-1} < \mathcal{N} \leq \bar{N}_j$ and $\mathcal{N}(p; \theta_0) = \int_{\theta_0}^1 n^*(\theta; p) g(\theta) d\theta$ decreasing in p , the unconstrained FOC price p^{FOC} is only *locally* optimal within a tier. The global optimum is obtained by comparing $\Pi(p^{FOC})$ to Π^* evaluated at *higher threshold* prices $\hat{p}_j : \hat{p}_j > p^{FOC}$ that solve $\mathcal{N}(\hat{p}_j; \theta_0) = \bar{N}_j$ (i.e., the smallest p that keeps expected usage within a lower cost tier j). Hence a different optimal price from p^{FOC} arises whenever $\Pi(\hat{p}_j) > \Pi(p^{FOC})$, because increasing p reduces $\mathcal{N}$ just enough to avoid the discrete cost jump $K_{j+1} - K_j$, which can dominate the marginal loss in gross profit along the concave $\tilde{\Pi}(p)$.
- **Platform Flywheel:** Expanding the supplier pool in a given category (via search and onboarding) creates a *participation gain* that, under Bulow–Klemperer, dominates the incremental gains from fine-tuning the auction format at a fixed supplier count. In our model, adding suppliers lowers expected procurement payments in category ω , raising effective per-requisition value from α to $\alpha' = \alpha + \lambda_\omega \delta_m^\omega$, which in turn increases fully optimized platform profit Π^* . Strategically, Fairmarkit should prioritize supplier recruitment

in **high-coverage categories** (large, type-invariant λ_ω) where **onboarding is scalable/low-cost** so that value creation scales across the customer base and can be monetized through a higher two-part tariff.

6 Forward-Looking Perspective: Sustaining Competitive Advantage

Fairmarkit's long-term viability and the sustainability of its competitive advantage against large incumbents and direct rivals require reinforcing the platform's core strategic defenses: maximizing value generation (α) and creating powerful customer lock-in.

6.1 Reinforcing the Core Advantage (α -Value)

Fairmarkit's per-requisition value parameter α is the engine of its success and the primary source of competitive advantage. As derived in the quantitative model, maximizing α requires a dual focus on minimizing process friction and maximizing competitive intensity. The forward-looking strategy should focus on the following operational levers:

- **Process and Search Time Savings ($\bar{h} - \tilde{h}$):** Continued AI development must drive the automated cycle time ($\tilde{h}$) towards zero. The existing platform offers massive cycle time compression, reducing processing hurdles (headcount to support the process) and search time per transaction. Future AI iterations must eliminate even more human intervention to keep the process savings term ($\bar{h} - \tilde{h}$) maximized while keeping equivalent quality of procurement choice.
- **Price Savings (s^{decl}):** The Bulow–Klemperer logic from Section 5.2 links supplier expansion in a given category ω to higher effective value creation: adding suppliers lowers expected procurement payments for that category and shifts the per-requisition parameter from α to $\alpha' = \alpha + \lambda_\omega \delta_m^\omega$. Under revenue submodularity, these savings exhibit diminishing marginal returns as the supplier count m grows. Supplier recruitment is privately justified as long as the incremental gain in *fully optimized* profit from adding the $(m+1)$ -st supplier, $\Delta\Pi_m := \Pi_{m+1}^* - \Pi_m^*$ (with $\Pi_m^* := \max_{(\theta_0, p)} \Pi_m(\theta_0, p)$), equals (or exceeds) the marginal search/onboarding cost $T_{m,\omega}$, i.e. $\Delta\Pi_m \geq T_{m,\omega}$, with $\Delta\Pi_m$ given in (18).

6.2 The Three Layers of Customer Retention (Appropriability and Lock-in)

Based on the nature of Fairmarkit's technology, specifically its deep integration with customer ERP systems and reliance on proprietary AI data, we can infer that customer retention is heavily influenced by multiple *lock-in mechanisms*. While these mechanisms were not explicitly quantified in the preceding sections, their effect on increasing switching costs is a critical component for the long-term sustainability and appropriability of Fairmarkit's business model.

The complexity of switching platforms creates high retention rates, securing the appropriability of Fairmarkit's business idea. The three key lock-in mechanisms can be conceptually ranked by their strategic importance and quantitative impact on switching cost in the following table (Table 3):

TABLE 3
Ranking of Fairmarkit's Customer Lock-in Mechanisms

Lock-in Mechanism	Description	Impact on Switching Cost
1. Data & Learning	The AI builds a proprietary, non-transferable data set unique to the customer (supplier responsiveness, pricing).	Highest / Ongoing: The value is the <i>future</i> loss of optimal bidding intelligence. It increases buyer value (ΔV) with every transaction.
2. Technical & Integration	Deep, bi-directional integration with core ERP/P2P systems (<i>SAP</i> , <i>Oracle</i> , <i>Coupa</i>).	High / Fixed Cost: Represents the complex IT project cost to <i>de-integrate</i> and <i>re-integrate</i> a new solution, creating operational risk.
3. Procedural & Organizational	Internal procurement teams are re-trained on the AI-driven workflow, making reversion to manual processes (emails, Excel) unviable.	Medium / Operational Risk: The cost of retraining staff and the efficiency loss of returning to the manual status quo (cycle time of 60 minutes vs. 3 minutes).

While the *Technical* lock-in presents a large, fixed-sum switching cost, the *Data & Learning* lock-in is the **most valuable long-term defense**. It increases the buyer's value (ΔV) with every transaction, making the platform progressively more indispensable and non-substitutable.

6.3 Sustaining Competitive Advantage Through Network Effects

The combination of high α -value and multi-layered lock-in mechanisms drives powerful, compounding **two-sided network effects**, which are Fairmarkit's ultimate forward defense. The virtuous cycle functions as a "flywheel":

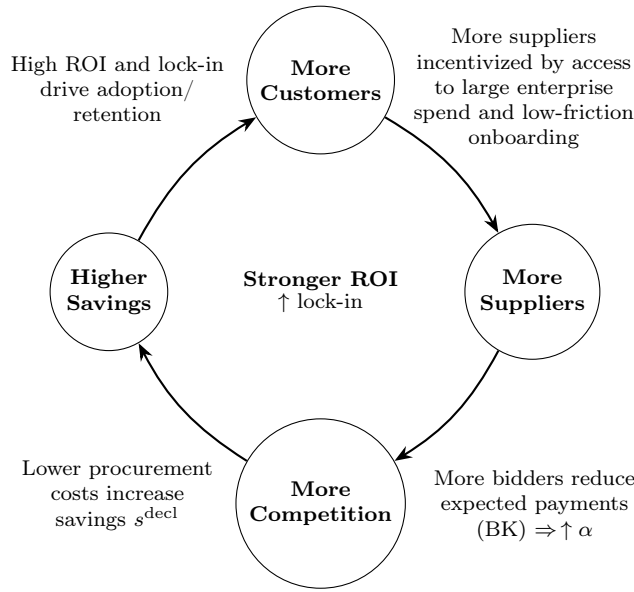


FIGURE 6

Virtuous flywheel cycle linking adoption, supplier depth, and realized savings.

This mechanism is further amplified by a **data network effect**, where global intelligence from aggregated, anonymized data allows the AI to identify trends and benchmark suppliers, making the platform smarter and more valuable for every user. By maximizing α and reinforcing these lock-in layers, Fairmarkit ensures its competitive advantage remains sustainable over the long term.

7 Conclusion & Future Extensions

This report implies three actionable levers for Fairmarkit's growth strategy. First, the optimal tariff should be designed for **selective coverage**: target customers above the cutoff type θ_0^* to protect margins while scaling adoption.

Second, pricing should rely on a **positive marginal price** alongside the access fee to manage shift to higher cost tiers given by higher requisition intensity.

Third, supplier strategy is an explicit investment decision: expand the supplier pool only while the **incremental gain in fully optimized profit** from higher auction effectiveness satisfies $\Delta \Pi_m^*(\theta_0^*) = T_{m,\omega}$. Overall, Fairmarkit should coordinate pricing, customer coverage, and supplier recruitment to **increase value creation per requisition** while keeping operational and search costs under control.

Several extensions could strengthen the analysis. A first direction is to incorporate **multi-homing** and **active competition from ERP bundles** to characterize optimal pricing when outside options are endogenous and competitive pressure directly constrains extractable surplus. Second, **distributional robustness** could be extended. Building on Sec. 5.1, one could calibrate the ambiguity radius ε from data (e.g., via KS confidence bands) and quantify how the least-favorable tail mass and conditional mean shift the optimal (θ_0, p) and expected profit, thereby providing an explicit worst-case performance guarantee over $\mathcal{G}_\varepsilon$.

Finally, the model could be extended to allow **more than one two-part tariff** (a menu of contracts), enabling additional screening and potentially improving profitability when customer heterogeneity is sufficiently rich and implementation constraints allows tiered offerings.

Moreover, the reverse-auction model can be extended beyond the i.i.d. benchmark. Under *independent but asymmetric* supplier cost distributions, Bulow–Klemperer-type results generally weaken from a dominance statement to an *approximation guarantee* relative to the optimal mechanism [29, 30]. In that setting, a natural next step is to quantify the additional implementation complexity of optimal mechanism design against the incremental profit gains from recruiting an additional supplier, using the same **total incremental optimal profit** metric developed in this report.

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